

Project Initialization and planning phase

Date	26/06/2026
Team Id	
Project Name	Smart Lender
Maximum Marks	5 Marks

Project Proposal Report

The proposed solution is **Smart Lender**, a Machine Learning-based web application that predicts the creditworthiness of loan applicants. By analyzing applicant details such as income, loan amount, credit history, and other relevant features, the system provides an instant prediction of loan approval or rejection. This solution helps financial institutions reduce manual effort, improve decision-making, and speed up the loan evaluation process.

1. Project Overview

Overview / Scope Description

Overview Smart Lender is a Machine Learning-based web application that predicts whether a loan application is likely to be approved or rejected based on applicant information.

Scope The project focuses on collecting applicant details, validating the input, processing the data through a trained Machine Learning model, and displaying the prediction result through a web interface.

2. Problem Statement

Description

Traditional loan approval involves manual verification and lengthy processing, which can lead to delays and inconsistent decisions.

Impact

Increases processing time, operational cost, and customer waiting time while reducing decision-making efficiency.

3. Proposed Solution

Approach

Develop a Flask-based web application integrated with a Random Forest Machine Learning model to analyze applicant information and predict loan approval status.

Key Features

- User-friendly web interface
- Input validation
- Machine Learning-based prediction
- Instant loan approval/rejection result
- Fast and accurate decision support
- Easy to maintain and extend

4. Resource Requirements

Resource	Description	Specification
Hardware	Computer/Laptop used for model development and application execution.	Intel Core i3 or above, 4 GB RAM (minimum), 20 GB free storage.
Software	Tools and technologies required to build and run the application.	Windows 10/11, Python 3.x, Flask, Jupyter Notebook, Visual Studio Code, Git.
Data	Loan applicant dataset used for training and testing the Machine Learning model.	Loan Prediction Dataset containing applicant income, loan amount, credit history, education, property area, and other relevant features.